From the Hessian to the Covariance: Generalising the CBMR standard-error step

Strategies, proofs, speed-ups, and what it costs in memory

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Abstract

A companion note derives the CBMR observed information in closed form for all three of NiMARE's observation distributions. That removes the Hessian as a bottleneck and immediately exposes the next one: `CBMRModel.covariance` spends its remaining time in two dense LAPACK calls, a singular value decomposition for a diagnostic condition number and an LU inverse. At the 21-group shape the SVD alone costs 163 s against the 1.6 s the Hessian now takes.

This note treats those two calls. We prove five propositions — structural separation, the blockwise inverse, the bordered (Schur) inverse, the block-diagonal spectrum, and the reduction of the condition number to a symmetric eigenproblem — and close each one symbolically with `sympy`, exactly as the companion note does for the Hessians. We then establish which designs each applies to and verify the resulting dispatch over the same nine-design, three-distribution suite. One candidate is rejected on measurement rather than on theory: LAPACK's `pocon` condition estimator is 18–264× off the true 2-norm value here, comfortably inside its worst-case bound and comfortably too far to report to a user. What generalises without any hypothesis on the design is simpler — the information matrix is symmetric, so a symmetric eigensolver replaces the SVD exactly.

We then measure memory, which the earlier work had not. On the real 21-group diagnosis pool (8,420 coefficients, 421 bases, 21,789 voxels, 8,178 experiments) a complete run — load, fit, covariance, and twenty one-versus-rest contrasts — peaks at 2.67 GB and takes 44 s, against 5.49 GB and 769 s for the chunked-autodiff path and a 28.7 GB intermediate for stock NiMARE, which is why the model could not be fitted at all.

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1 The problem

Let $H \in \mathbb{R}^{n \times n}$ be the observed information over the $n = CK + Q$ regression coefficients. `CBMRModel.covariance` does two things with it:

1. computes $\kappa_2(H) = \sigma_{\max}/\sigma_{\min}$ with `np.linalg.cond`, purely to decide whether to warn that the standard errors are untrustworthy;
2. inverts it with `np.linalg.inv`.

`np.linalg.cond` forms a full singular value decomposition. At $n = 7{,}329$ that is 163 s, against 14 s for the inverse it is diagnosing and 1.6 s for the Hessian itself. The diagnostic costs an order of magnitude more than the quantity.

2 Four strategies

Write $L \in \mathbb{R}^{P \times C}$ for the loadings, $\operatorname{supp}(p) = \{c : L_{pc} \neq 0\}$, and let $\sim$ be the transitive closure of $c \sim c' \iff \exists p : \; c, c' \in \operatorname{supp}(p)$.

Proposition 1 (Structural separation)**.** *For every one of the three distributions, $H_{(ck),(c'k')} = 0$ whenever $c \not\sim c'$.*

Proof. The companion note shows each distribution's Hessian has $\beta\beta$ block $\sum_p L_{pc} L_{pc'} (B^\top \Sigma_p B)_{kk'}$. Every summand carries the factor $L_{pc} L_{pc'}$, which vanishes unless $c, c' \in \operatorname{supp}(p)$; if some p has both, then $c \sim c'$. □

That sparsity is the raw material here. It is a statement about the design, not the observation model, and the entries are structurally absent rather than merely small.

Proposition 2 (Block-diagonal spectrum)**.** *If $H = \operatorname{blkdiag}(H_1, \dots, H_M)$ then $\operatorname{spec}(H) = \bigsqcup_j \operatorname{spec}(H_j)$, and the singular values are likewise the multiset union. Hence $\kappa_2(H) = \max_j \sigma_{\max}(H_j) \big/ \min_j \sigma_{\min}(H_j)$, exactly.*

Proof. $H - \lambda I$ is block diagonal with blocks $H_j - \lambda I$, and the determinant of a block diagonal matrix is the product of its blocks' determinants, so the characteristic polynomial factorises. Applying the same to $H^\top H$ gives the singular values. □

Proposition 3 (Symmetry suffices)**.** *H is symmetric by construction. For symmetric H with eigenvalues λ_i the singular values are $|\lambda_i|$, so $\kappa_2(H) = \max_i |\lambda_i| / \min_i |\lambda_i|$ and a symmetric eigensolver may replace the SVD. No hypothesis on the design, and no positive definiteness.*

Proof. Write $H = V \Lambda V^\top$ with V orthogonal; then $H^\top H = V \Lambda^2 V^\top$, so the singular values — the nonnegative square roots of the eigenvalues of $H^\top H$ — are $|\lambda_i|$. □

Lemma 4 (Blockwise inverse)**.** *If $H = \operatorname{blkdiag}(H_1, \dots, H_M)$ with each H_j nonsingular, then $H^{-1} = \operatorname{blkdiag}(H_1^{-1}, \dots, H_M^{-1})$.*

Proof. The product $\mathrm{blkdiag}(H_j)\,\mathrm{blkdiag}(H_j^{-1})$ is block diagonal with j-th block $H_jH_j^{-1} = I$, so it is the identity; inverses are unique. □

Proposition 5 (Bordered inverse)**.** *Suppose the spatial columns separate but $Q > 0$, so that*

$$H = \begin{pmatrix} D & C \\ C^\top & E \end{pmatrix}, \qquad D = \mathrm{blkdiag}(D_1, \ldots, D_M) \in \mathbb{R}^{CK \times CK}, \quad C \in \mathbb{R}^{CK \times Q}.$$

If D and $\Sigma = E - C^\top D^{-1} C$ are nonsingular then

$$H^{-1} = \begin{pmatrix} D^{-1} + D^{-1}C\Sigma^{-1}C^\top D^{-1} & -D^{-1}C\Sigma^{-1} \\ -\Sigma^{-1}C^\top D^{-1} & \Sigma^{-1} \end{pmatrix},$$

at cost $O(\sum_j |D_j|^3) + O(Q^3)$ for the factorisations rather than $O((CK+Q)^3)$.

Proof. Standard block inversion; verify by multiplying out $HH^{-1} = I$ blockwise using $E - C^\top D^{-1} C = \Sigma$. □

Proposition 6 (One factorisation for both)**.** *If $H \succ 0$ then a single Cholesky factorisation $H = LL^\top$ serves twice: `pocon` returns an estimate of $\kappa_1(H)$ in $O(n^2)$ from L, and `potri` returns H^{-1} in $O(n^3/3)$ from the same L. Moreover*

$$n^{-1}\kappa_1(H) \;\le\; \kappa_2(H) \;\le\; n\,\kappa_1(H). \tag{1}$$

Proof. The routines are standard. For (1), apply $n^{-1/2}\|M\|_1 \le \|M\|_2 \le n^{1/2}\|M\|_1$ to $M = H$ and $M = H^{-1}$ and multiply. □

3 What the measurement rejected

Proposition 6 is by far the fastest route and (1) makes it look safe: a factor of n is four of the sixteen available decimal digits. Measured on the three structures of Table 1, the `pocon` estimate reports 90.09 against a true κ_2 of 5.0243, 2.42×10^5 against 1657.7, and 1312.4 against 4.9796 — errors of 18×, 146× and 264×.

Remark 7. Every one of those is inside the bound, which is the point. A bound that admits a 264× error admits behaviour that cannot be shown to a user as a condition number. A fit at $\kappa_2 = 10^{14}$ — badly conditioned but genuinely invertible — would be reported near 10^{16} and trip a warning saying its standard errors must not be trusted.

The resolution is to split the two uses of the factorisation. The Cholesky is kept for the *inverse*, where it agrees with `np.linalg.inv` to 1.8×10^{-15} and runs 2–3× faster. It is discarded for the *condition number*, which comes instead from Proposition 3: exact, fully general, and still 3–4× faster than the SVD.

4 The dispatch, and its generality

1. **blockwise** — spatial columns separate, $Q = 0$. Condition number by Proposition 2, inverse blockwise.
2. **bordered** — spatial columns separate, $Q > 0$. Inverse by Proposition 5; the condition number cannot follow, because the spectrum of a bordered matrix is not the union of its parts, so it falls back to Proposition 3.
3. **Cholesky** — no exploitable structure but $H \succ 0$. Condition number by Proposition 3, inverse by Cholesky.

Table 1: Condition number and inverse on synthetic matrices at the 21-group shape, over the three structures real designs produce. "dense" is the stock path. All inverses agree with the dense one to $\leq 4.6 \times 10^{-15}$ relative. Rows marked $\star$ are what the dispatch adopts.

structure	strategy	κ reported	κ time	inverse time	total
block diagonal `~ s(a)`, 7329^2	dense (SVD + LU)	5.0243	157.2 s	14.2 s	171.4 s
	`eigvalsh` + Cholesky	5.0243	36.5 s	7.6 s	44.1 s
	Cholesky + `pocon`	90.093	2.9 s	5.5 s	8.4 s
	$\star$ blockwise	5.0243	0.4 s	0.2 s	**0.7** s
bordered `~ s(a) + x * z`, 7331^2	dense (SVD + LU)	1657.7	169.0 s	20.7 s	189.6 s
	$\star$ `eigvalsh` + Schur	1657.7	48.5 s	4.1 s	**52.7** s
	Cholesky + `pocon`	2.42×10^5	4.7 s	8.7 s	13.3 s
	blockwise	*not applicable — the matrix does not separate*			
fully coupled `~ sz(a) + sz(b)`, 7329^2	dense (SVD + LU)	4.9796	161.8 s	19.8 s	181.7 s
	$\star$ `eigvalsh` + Cholesky	4.9796	52.1 s	15.8 s	**67.8** s
	Cholesky + `pocon`	1312.4	4.4 s	8.6 s	13.0 s

4. **dense** — $H \not\succ 0$; the Cholesky declines and the general inverse runs.

Proposition 8 (The dispatch is well defined)**.** *On every branch the dispatch returns H^{-1} and the exact $\kappa_2(H)$, so the rung taken affects only cost. Moreover rung 2 reduces to rung 1 when $C = 0$.*

Proof. Rung 1 returns H^{-1} by Lemma 4 and $\kappa_2(H)$ by Proposition 2; rung 2 by Proposition 5 and Proposition 3; rung 3 by uniqueness of the Cholesky inverse and Proposition 3; rung 4 by definition. For the reduction, set $C = 0$ in Proposition 5: then $\Sigma = E$, the off-diagonal blocks of the stated inverse vanish and the leading block is D^{-1}, which is Lemma 4. □

Every rung therefore reports the *same* condition number, to the last printed digit. Structural claims are checked against the matrix in hand, never assumed: a design that ought to separate but whose information matrix has an entry outside the implied blocks is inverted densely, with a warning.

Table 2 verifies the dispatch over the nine-design, three-distribution suite. All four rungs fire; every one agrees with the dense inverse to within a tolerance scaled by the matrix's own conditioning. The important negative result is the bordered rung: a global moderator destroys block diagonality outright — one block where there were twenty — while leaving the spatial columns separable underneath, so the inverse is recovered even though the classification that motivated it is lost.

5 Equivalence

Two independent checks, matching the companion note's structure.

5.1 Symbolic

`scripts/symbolic_covariance.py` restates each proposition over symbols and closes it with `sympy`. Every residual is exactly 0; nothing numerical is involved. (The script runs structure, then inverses, then spectra; the table is ordered by proposition instead.)

Table 2: The dispatch over every design the CBMR grammar produces, crossed with all three distributions. "blocks" counts the components of $\sim$; "rung" is what that buys. Relative error is against `np.linalg.inv` of the same matrix, worst over the distributions that accept the design.

design	formula	blocks	rung	κ_2	inverse rel. error
cell-means factor	`~ s(a)`	4	blockwise	4.8×10^4	4.0×10^{-13}
full interaction	`~ s(a:b)`	8	blockwise	4.7×10^4	1.6×10^{-12}
additive main effects	`~ sz(a) + sz(b)`	1	Cholesky	1.9×10^5	1.2×10^{-12}
main effects + interaction	`~ sz(a) + sz(b) + sz(a:b)`	1	dense[†]	4.1×10^{20}	—
one global moderator	`~ s(a) + x`	1	bordered	2.5×10^7	9.0×10^{-13}
global moderator interaction	`~ s(a) + x * z`	1	bordered	2.6×10^7	8.3×10^{-13}
voxelwise moderator, continuous	`~ s(a) + s(x)`	1	Cholesky	2.3×10^5	6.4×10^{-13}
voxelwise moderator, coarse	`~ s(a) + s(w)`	1	Cholesky	6.2×10^6	7.6×10^{-12}
everything at once	`~ sz(a) + sz(b) + s(w) + x * z`	1	Cholesky	5.1×10^7	3.3×10^{-11}

[†] Rank deficient at this sample size, so the Cholesky declines and the general inverse runs. No inverse of this matrix is trustworthy by any route, so no comparison is made; the test asserts the singularity instead.

claim	residual checked	result
Prop. 1	cross block of two columns sharing no pattern	0
Prop. 2	$\det(\mathrm{blkdiag}(A, B) - \lambda I) - \det(A - \lambda I)\det(B - \lambda I)$	0
Prop. 3	$H^\top H - H^2$ for symmetric H	0
	$\det(H^2 - \lambda^2 I) - \det(H - \lambda I)\det(H + \lambda I)$	0
Lem. 4	$\mathrm{blkdiag}(A, B)^{-1} - \mathrm{blkdiag}(A^{-1}, B^{-1})$	0
Prop. 5	$D\,\mathrm{TL} + C\,\mathrm{BL} - I_m$	0
	$D\,\mathrm{TR} + C\,\mathrm{BR}$	0
	$C^\top \mathrm{TL} + E\,\mathrm{BL}$	0
	$C^\top \mathrm{TR} + E\,\mathrm{BR} - I_q$	0

Remark 9 (Why Proposition 5 is proved over matrix symbols)**.** Filling in a 4×4 symbolic D^{-1} inside a 6×6 product makes `simplify` intractable — the attempt does not terminate. It also proves less. The identity needs only that D and Σ are invertible and that the shapes conform, so D^{-1} and Σ^{-1} are left formal and the sole rewrites used are $DD^{-1} = I$ and, through $E = \Sigma + C^\top D^{-1} C$, the definition of the Schur complement. The result is a proof for arbitrary conformable blocks rather than for one filled-in instance, and Lemma 4 is what licenses computing that D^{-1} blockwise.

5.2 Numerical

`test_cbmr_analytic.py` applies all four rungs to *the same* matrix rather than to four separate fits, so nothing about a fit can mask a disagreement: a block diagonal 48×48 and the same matrix with a 2-column border welded on. Blockwise, Cholesky and both routes to the condition number match the dense ones to 10^{-10}; setting the border to zero returns the Schur complement to the blockwise inverse exactly, which is Proposition 8's reduction claim in arithmetic. `test_cbmr_designs.py` then exercises the dispatch on real fits, over the suite of Table 2.

6 Memory

Speed was measured first because it was the visible failure. Memory is the one that decides whether the model can be fitted at all.

6.1 Where it goes

Three allocations dominate, all $O(n^2)$: the information matrix, the covariance returned to the caller, and whatever working copy the chosen strategy makes. At the real 21-group shape $n = 8{,}420$, so each is 0.53 GB. Everything else — the sparse foci, the $(V \times K)$ basis, the L-BFGS history, the per-pattern log-intensity — is an order of magnitude smaller.

Remark 10 (A saving that is not there)**.** `np.zeros` is lazily backed, and the blockwise inverse writes only the diagonal blocks — 19 MB of a 400 MB array at the synthetic shape. It is tempting to conclude that a block-diagonal covariance is nearly free in resident memory. Measured, it is not: writing the diagonal blocks moves resident memory by +0.399 GB, essentially the whole array, because the blocks are strided across every row and the fault granularity is coarser than the writes. Reads genuinely do not fault — two full passes over the array cost +0.000 GB — but that is no help, since the array is fully resident by then. Budget n^2 words, not the blocks.

6.2 Measured, per call

Table 3 reports peak resident memory *during* the `covariance()` call, sampled at 100 Hz from `/proc/self/statm`. `ru_maxrss` is the wrong instrument here: it is a high-water mark for the whole process, so it reports whatever the setup allocated.

Table 3: Peak resident memory during one `covariance()` call, one process per row. "matrix" is n^2 words, the unavoidable floor. The last column is the intermediate stock `torch.func.hessian` would require, which is why none of these shapes runs on it.

shape	n	rung	time	peak Δ	matrix	stock needs
diagnosis, 10 groups	4,100	blockwise	1.1 s	0.18 GB	0.13 GB	4.9 GB
task, 10 groups	4,270	blockwise	1.8 s	0.19 GB	0.14 GB	7.7 GB
diagnosis, 21 groups	7,329	blockwise	1.3 s	0.48 GB	0.40 GB	9.7 GB
diagnosis, 21 groups + moderator	7,330	bordered	40.6 s	1.28 GB	0.40 GB	9.7 GB

The blockwise rung costs about what the matrix costs. The bordered rung costs roughly three times that, and is 30× slower: it pays for a full symmetric eigendecomposition of the whole matrix (Proposition 3 cannot be localised to the blocks) and for a dense D^{-1} alongside the result. It is still a large improvement on the dense path, but it is the expensive rung, and a design that avoids moderators avoids it.

6.3 Measured, end to end

Table 4 is the number that answers the practical question. The run is the real 21-group diagnosis pool — every category with at least 200 studies, excluding Healthy — at 4 mm with `spline_spacing=5`: 20 groups after pooling, 8,178 experiments, 8,420 coefficients, 421 bases, 21,789 analysis voxels. It covers everything a user does: load, fit, covariance, and twenty one-versus-rest contrasts.

17× faster and half the memory. The contrast stage improves by more than the memoisation alone predicts; part of that gap is likely contention on the shared machine and should not be read as a 25× effect of this change.

6.4 Will it run on a laptop?

At 2.67 GB peak, the complete 21-group analysis fits on a 10–12 GB machine with room to spare — roughly a quarter of available RAM, leaving the operating system and everything else untouched. The 10-group diagnosis and task models are smaller still. Three caveats:

Table 4: Complete 21-group run, real data, cumulative peak `ru_maxrss` at each stage. "chunked" is the previous `jacrev` path with the dense SVD and LU; "closed form" is the closed-form Hessian with the dispatch of Section 4. Stock NiMARE does not appear because it does not complete: `torch.func.hessian` requires a 28.7 GB intermediate at this shape.

	chunked autodiff		closed form	
stage	time	peak	time	peak
load studyset	1.8 s	0.74 GB	1.8 s	0.74 GB
fit + covariance	398.7 s	4.40 GB	27.4 s	2.07 GB
20 one-versus-rest contrasts	368.8 s	5.49 GB	14.7 s	2.67 GB
total	**769** s	**5.49** GB	**44** s	**2.67** GB

- Memory grows as $n^2 = (CK + Q)^2$, and C is the group count. Doubling the groups quadruples the covariance: 40 groups at 421 bases gives $n = 16{,}840$ and 2.11 GB per matrix. Extrapolating the measured ratio of peak to matrix size on the blockwise rung ($\approx$1.2), that is roughly 6–8 GB peak — still inside a 12 GB machine, but no longer with room to spare. This is an extrapolation, not a measurement. It is the ceiling to watch, and `spline_spacing` moves it as directly as the group count does.
- Adding a global moderator moves the fit onto the bordered rung, which measured 2.7× the memory and 30× the time of the blockwise one.
- The covariance is returned dense because `contrasts.py` indexes it densely. That 0.53 GB is the floor; removing it means changing NiMARE.

7 Identification

At the real shape there are 8,420 coefficients and 8,178 experiments, which invites the conclusion that the model has more parameters than data. It does not, and the reason is worth recording because the reassuring version of the argument is also wrong.

CBMR's likelihood is not over experiment rows. It is over voxel-level foci counts, summarised per pattern, so there are $P \times V = 435{,}780$ count observations — 52 per coefficient. More to the point, the spatial factor term is block diagonal, so the model is not one problem with 8,420 parameters but twenty independent problems with 421 each, and each group's coefficients are identified by that group's own foci. Measured over the real pool, foci per coefficient runs from 8.1 (hearing loss, 221 studies, 3,406 foci) to 65.4 (Alzheimer's, 1,345 studies, 27,529 foci), and every per-group block has condition number between 3.8×10^4 and 3.0×10^5 — far from the $\epsilon^{-1} \approx 4.5 \times 10^{15}$ at which double precision gives out.

Remark 11 (The diagnostic does not diagnose this)**.** Under a canonical link the observed information equals the expected information, so for Poisson H is a function of the fitted intensity and the group sizes and not of the foci directly. A group can be far too thin for the asymptotics while still producing a perfectly invertible block. Fitting three groups of sizes $(40, 40, 40)$, $(40, 40, 10)$, $(40, 40, 4)$ and $(40, 40, 2)$, the thin group's condition number stays between 1.3×10^4 and 1.5×10^5 throughout — no trend, no warning — while its median coefficient standard error rises from 27.5 to 150.2. The standard errors do respond; the conditioning warning does not. It is the wrong instrument for detecting a data-starved group, and `incidence_threshold` together with the per-term parameter budget from `describe_terms()` is the right one.

8 Generated proof fragments

Everything below is written by `run_proofs.py` from the same sympy objects the proofs check. Nothing here is transcribed by hand, so the document cannot disagree with the verification that backs it.

8.1 Poisson

The log-likelihood on marginals, dropping the parameter-free $-\sum \log y!$, is $\log \mathcal{L} = \sum_v Y_v S_v + \sum_i y_i m_i - (\sum_v e^{S_v})T$ with $T = \sum_i e^{m_i}$. The first two terms are linear in the coefficients, so they differentiate away entirely; that is the whole of the data dependence.
The expressions the claims are formed from:

$$-\log \mathcal{L} = -Y_0\,(B_{00}\beta_0 + B_{01}\beta_1) - Y_1\,(B_{10}\beta_0 + B_{11}\beta_1) - Y_2\,(B_{20}\beta_0 + B_{21}\beta_1) - y_0\,(G_{00}\gamma_0 + G_{01}\gamma_1) - y_1\,(G_{10}\gamma_0 + G_1$$

Residuals, each asserted to vanish:

claim	residual	result
d2/dbeta0 dbeta0	$\partial^2_{\beta_0\beta_0} - T\,(B^\top \mathrm{diag}(u) B)_{00}$	0
d2/dbeta0 dbeta1	$\partial^2_{\beta_0\beta_1} - T\,(B^\top \mathrm{diag}(u) B)_{01}$	0
d2/dbeta1 dbeta0	$\partial^2_{\beta_1\beta_0} - T\,(B^\top \mathrm{diag}(u) B)_{10}$	0
d2/dbeta1 dbeta1	$\partial^2_{\beta_1\beta_1} - T\,(B^\top \mathrm{diag}(u) B)_{11}$	0
d2/dbeta0 dgamma0	$\partial^2_{\beta_0\gamma_0} - a_0 U_0$	0
d2/dbeta0 dgamma1	$\partial^2_{\beta_0\gamma_1} - a_0 U_1$	0
d2/dbeta1 dgamma0	$\partial^2_{\beta_1\gamma_0} - a_1 U_0$	0
d2/dbeta1 dgamma1	$\partial^2_{\beta_1\gamma_1} - a_1 U_1$	0
d2/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0} - E\sum_i e_i G_{i0} G_{i0}$	0
d2/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1} - E\sum_i e_i G_{i0} G_{i1}$	0
d2/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0} - E\sum_i e_i G_{i1} G_{i0}$	0
d2/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1} - E\sum_i e_i G_{i1} G_{i1}$	0
Hessian contains no foci symbol	$\{Y_v, y_i\} \cap \mathrm{free}(H)$	0

Verdict: every residual is exactly zero.

8.2 Negative Binomial

NiMARE's negative binomial uses shape $r = s_1^2/(\theta s_2)$ and success probability $\pi_v = (1 + s_1/(\theta u_v s_2))^{-1}$. Writing $\eta = \log s_1$ and $\kappa = \log k$ with $k = \theta s_2/s_1^2$, the voxel mean is $\mu_v = e^\eta u_v$, the size is $1/k$, and each voxel's summand depends on (η, κ) and S_v alone. Below, $x_v = k\mu_v$.
The expressions the claims are formed from:

$$\lambda(\eta, \kappa, S_v) = \kappa y + y \log\left(e^S e^\eta\right) - \left(y + e^{-\kappa}\right) \log\left(e^S e^\eta e^\kappa + 1\right) + \mathrm{loggamma}\left(y + e^{-\kappa}\right) - \mathrm{loggamma}\left(e^{-\kappa}\right)$$

One voxel's log-likelihood, up to $-\log\Gamma(Y_v + 1)$.

$$G_1 = y - \left(\mathrm{polygamma}\left(0, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(0, e^{-\kappa}\right)\right) e^{-\kappa}$$

The count term of λ_κ. polygamma$(0, \cdot)$ is the digamma function.

$$G_2 = \left(\mathrm{polygamma}\left(0, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(0, e^{-\kappa}\right)\right) e^{-\kappa} + \left(\mathrm{polygamma}\left(1, y + e^{-\kappa}\right) - \mathrm{polygamma}\left(1, e^{-\kappa}\right)\right) e$$

The count term of $\lambda_{\kappa\kappa}$.

$$g(x_v) = -\frac{1}{x_v + 1} + \frac{\log(x_v + 1)}{x_v}$$

Residuals, each asserted to vanish:

claim	residual	result
lambda(eta,kappa,S) equals NiMARE's summand	$\lambda\big\|_{\eta,\kappa} - \big(\log \mathcal{L}_v + \log\Gamma(Y_v + 1)\big)$	0
d lambda/dS = d lambda/deta	$\partial_{S_v}\lambda - \partial_\eta\lambda$	0
lambda_eta	$\lambda_\eta - \frac{Y_v - \mu_v}{1+x_v}$	0
lambda_eta_eta	$\lambda_{\eta\eta} + \frac{\mu_v + Y_v x_v}{(1+x_v)^2}$	0
lambda_eta_kappa	$\lambda_{\eta\kappa} + \frac{(Y_v - \mu_v)x_v}{(1+x_v)^2}$	0
lambda_kappa	λ_κ as implemented	0
lambda_kappa_kappa	$\lambda_{\kappa\kappa}$ as implemented	0
count sum in lambda_kappa, y=0	$G_1(Y_v = 0) - \sum_{j<0} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=0	$G_2(Y_v = 0) - \sum_{j<0} \frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=1	$G_1(Y_v = 1) - \sum_{j<1} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=1	$G_2(Y_v = 1) - \sum_{j<1} \frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=2	$G_1(Y_v = 2) - \sum_{j<2} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=2	$G_2(Y_v = 2) - \sum_{j<2} \frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=3	$G_1(Y_v = 3) - \sum_{j<3} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=3	$G_2(Y_v = 3) - \sum_{j<3} \frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=4	$G_1(Y_v = 4) - \sum_{j<4} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=4	$G_2(Y_v = 4) - \sum_{j<4} \frac{jk}{(1+jk)^2}$	0
count sum in lambda_kappa, y=5	$G_1(Y_v = 5) - \sum_{j<5} \frac{jk}{1+jk}$	0
count sum in lambda_kappa_kappa, y=5	$G_2(Y_v = 5) - \sum_{j<5} \frac{jk}{(1+jk)^2}$	0
gap series to xˆ10	$\left[\frac{\log(1+x)}{x} - \frac{1}{1+x}\right]_{x^1 0} - \sum_{n=1}^{10} \frac{(-1)^{n+1} n x^n}{n+1}$	0
deta/dgamma0	$\partial_{\gamma_0}\eta - \ell_0$	0
dkappa/dgamma0	$\partial_{\gamma_0}\kappa - (h_0 - 2\ell_0)$	0
d2eta/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}\eta - \ell^{(2)}_{00}$	0
d2kappa/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}\kappa - (h^{(2)}_{00} - 2\ell^{(2)}_{00})$	0
d2eta/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}\eta - \ell^{(2)}_{01}$	0
d2kappa/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}\kappa - (h^{(2)}_{01} - 2\ell^{(2)}_{01})$	0
deta/dgamma1	$\partial_{\gamma_1}\eta - \ell_1$	0
dkappa/dgamma1	$\partial_{\gamma_1}\kappa - (h_1 - 2\ell_1)$	0
d2eta/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0}\eta - \ell^{(2)}_{10}$	0
d2kappa/dgamma1 dgamma0	$\partial^2_{\gamma_1\gamma_0}\kappa - (h^{(2)}_{10} - 2\ell^{(2)}_{10})$	0
d2eta/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}\eta - \ell^{(2)}_{11}$	0
d2kappa/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}\kappa - (h^{(2)}_{11} - 2\ell^{(2)}_{11})$	0

Verdict: every residual is exactly zero.

8.3 Clustered Negative Binomial

The latent effect is a property of the experiment rather than of the voxel, so the spatial coefficients reach the likelihood only through the scalar $E = \sum_v u_v$. Writing $\Gamma(E) = \sum_i (y_i + \nu)\log(Ee_i + \nu)$, the second differential in S is $\Gamma''(E)u_v u_{v'} + \Gamma'(E)u_v\delta_{vv'}$ — diagonal plus rank one.
The expressions the claims are formed from:

$$-\log \mathcal{L}_p = -Y_0\left(B_{00}\beta_0 + B_{01}\beta_1\right) - Y_1\left(B_{10}\beta_0 + B_{11}\beta_1\right) - Y_2\left(B_{20}\beta_0 + B_{21}\beta_1\right) - 2\nu\log(\nu) - y_0\left(G_{00}\gamma_0 + G_{01}\gamma_1\right) - y_1$$

$$g' = \frac{(\nu + y_0)\, e^{G_{00}\gamma_0 + G_{01}\gamma_1}}{\nu + \left(e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 + B_{21}\beta_1}\right) e^{G_{00}\gamma_0 + G_{01}\gamma_1}} + \frac{(\nu + y_1)\, e^{G_{10}\gamma_0 + G_{11}\gamma_1}}{\nu + (e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 +}}$$

$$g'' = -\frac{(\nu + y_0)\, e^{2G_{00}\gamma_0 + 2G_{01}\gamma_1}}{\left(\nu + \left(e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e^{B_{20}\beta_0 + B_{21}\beta_1}\right) e^{G_{00}\gamma_0 + G_{01}\gamma_1}\right)^2} - \frac{(\nu + y_1)\, e^{2G_{10}\gamma_0 +}}{(\nu + (e^{B_{00}\beta_0 + B_{01}\beta_1} + e^{B_{10}\beta_0 + B_{11}\beta_1} + e}$$

Residuals, each asserted to vanish:

claim	residual	result
d2/dbeta0 dbeta0	$\partial^2_{\beta_0\beta_0}$ − closed form	0
d2/dbeta0 dbeta1	$\partial^2_{\beta_0\beta_1}$ − closed form	0
d2/dbeta0 dgamma0	$\partial^2_{\beta_0\gamma_0}$ − closed form	0
d2/dbeta0 dgamma1	$\partial^2_{\beta_0\gamma_1}$ − closed form	0
d2/dbeta1 dbeta1	$\partial^2_{\beta_1\beta_1}$ − closed form	0
d2/dbeta1 dgamma0	$\partial^2_{\beta_1\gamma_0}$ − closed form	0
d2/dbeta1 dgamma1	$\partial^2_{\beta_1\gamma_1}$ − closed form	0
d2/dgamma0 dgamma0	$\partial^2_{\gamma_0\gamma_0}$ − closed form	0
d2/dgamma0 dgamma1	$\partial^2_{\gamma_0\gamma_1}$ − closed form	0
d2/dgamma1 dgamma1	$\partial^2_{\gamma_1\gamma_1}$ − closed form	0

Verdict: every residual is exactly zero.

8.4 Covariance strategies

Let $\mathrm{supp}(p) = \{c : L_{pc} \neq 0\}$ and let $\sim$ be the transitive closure of $c \sim c' \iff \exists p :\ c, c' \in \mathrm{supp}(p)$. Every distribution's Hessian has $\beta\beta$ block $\sum_p L_{pc} L_{pc'} (B^\top \Sigma_p B)_{kk'}$, so the factor $L_{pc} L_{pc'}$ decides which entries can be nonzero at all.
Residuals, each asserted to vanish:

claim	residual	result
columns sharing no pattern have a zero cross block	$H_{(0k),(2k')}$ with $0 \not\sim 2$	0
characteristic polynomial factorises over the blocks	$\det(\mathrm{blkdiag}(A, B) - \lambda I) - \det(A - \lambda I)\det(B - \lambda I)$	0
symmetric H satisfies H^T H = H^2	$H^\top H - H^2$	0
the singular values are the absolute eigenvalues	$\det(H^2 - \lambda^2 I) - \det(H - \lambda I)\det(H + \lambda I)$	0
blockwise inverse	$\mathrm{blkdiag}(A, B)^{-1} - \mathrm{blkdiag}(A^{-1}, B^{-1})$	0
D.TL + C.BL = I	$D\,\mathrm{TL} + C\,\mathrm{BL} - I_m$	0
D.TR + C.BR = 0	$D\,\mathrm{TR} + C\,\mathrm{BR}$	0
C^T.TL + E.BL = 0	$C^\top \mathrm{TL} + E\,\mathrm{BL}$	0
C^T.TR + E.BR = I	$C^\top \mathrm{TR} + E\,\mathrm{BR} - I_q$	0

Verdict: every residual is exactly zero.